CLIMATEWINS:
Weather Conditions & Climate Change

With Machine Learning

Presentation - 2025

AMRIT GILL

GIILLAMRIT03@OUTLOOK.COM

OBJECTIVES

ClimateWins is adopting Machine Learning to acheive following goals:

- Identify weather patterns outside the regional norm in Europe
- Determine if unusual weather patterns are increasing
- Generate possibilities for future weather conditions over the next 20 to 50 years based on current trends
- Determine the safest places for people to live in Europe over the next 25 to 50 years

THREE THOUGHT EXPRIMENTS

RANDOM FOREST

Focus on using ensemble models to analyze shifts in weather patterns based on historical data.

CNNs & RNNs

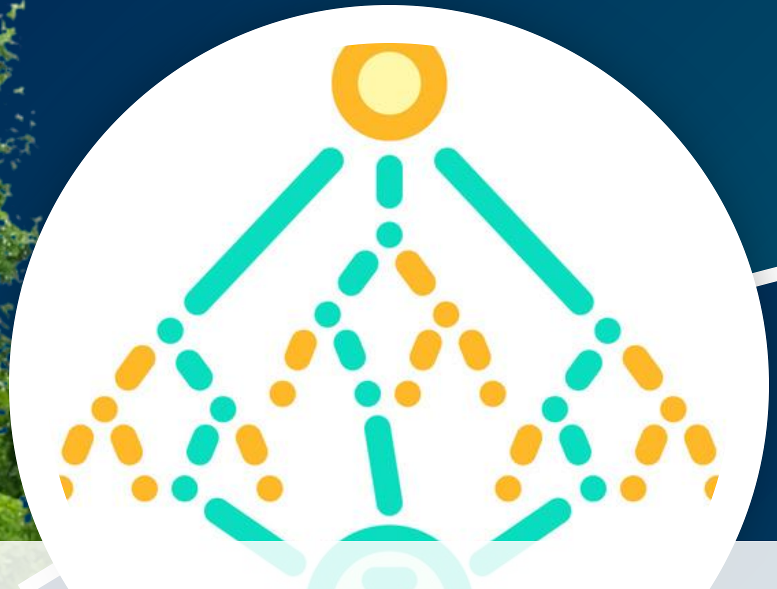
Utilize Convolutional Neural Networks (CNNs) to detect patterns in satellite images and radar data.

GANs

Involve two networks, a generator and a

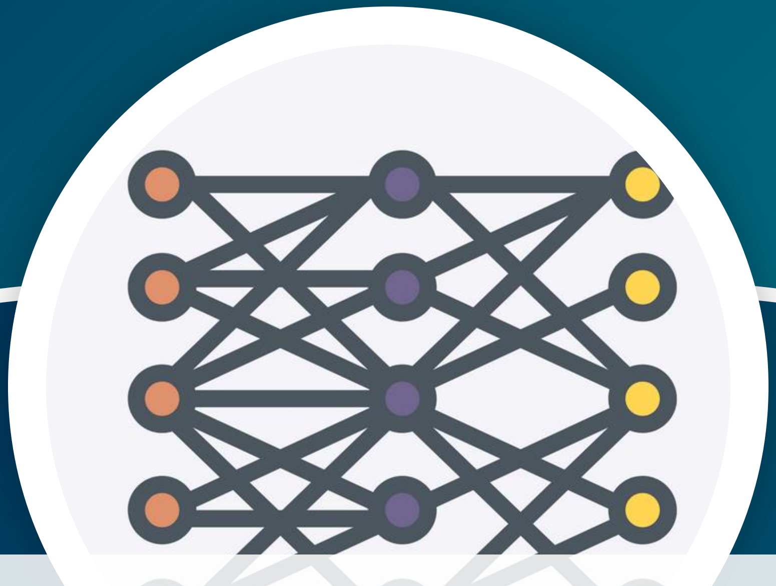
discriminator, competing to create realistic synthetic data.

How Three Thought Expriments Used:



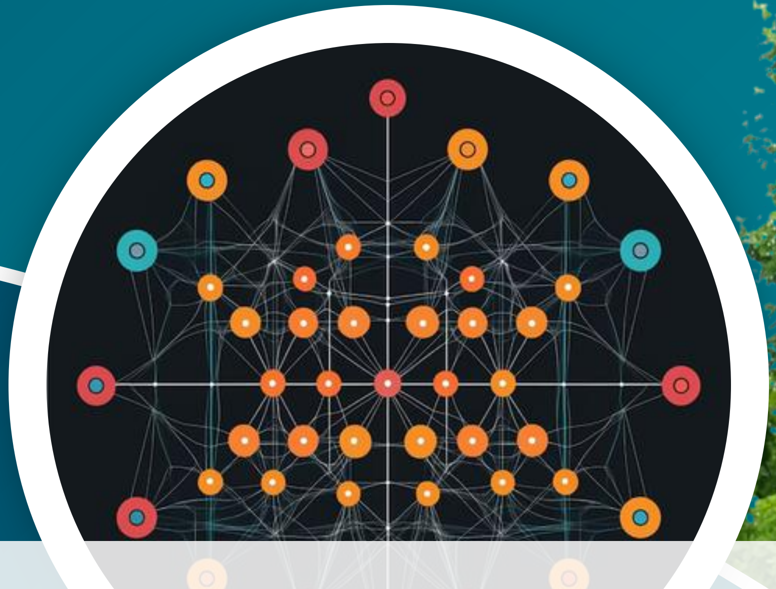
Random Forest

Can be used to predict weather analysis.
outcomes by analyzing multiple
environmental features and their
interactions.



CNNs & RNNs

Can analyze satellite images for weather pattern recognition, while RNNs can model time series data to forecast future weather conditions based on historical trends.



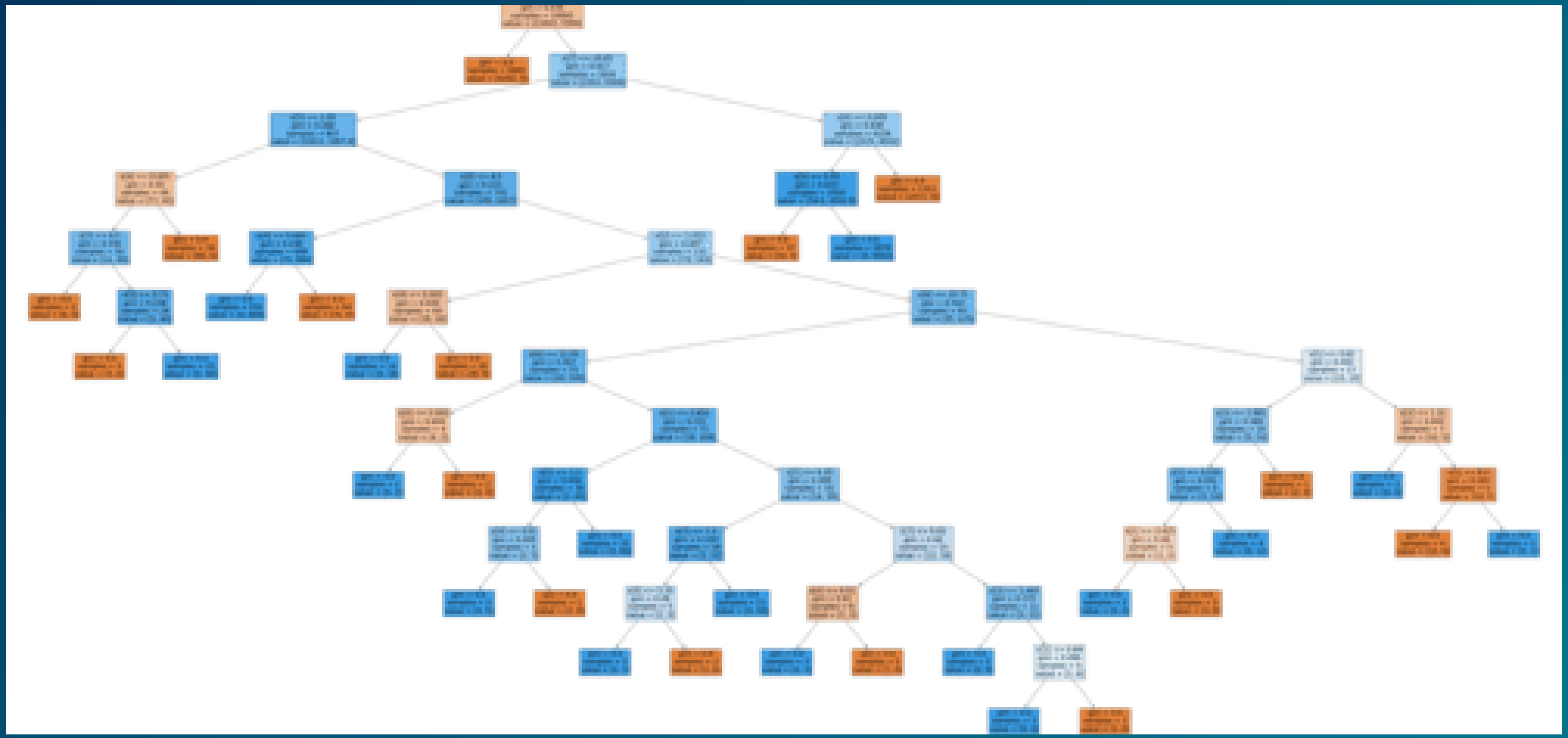
GANs

Can generate realistic synthetic weather data to augment datasets for training models, improving the robustness of predictions in weather analysis.

Machine Learning Model 1

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees using random subsets of data and features to improve predictive performance. By aggregating the outputs of these diverse trees—typically through majority voting for classification or averaging for regression—it enhances accuracy and reduces the risk of overfitting.

Random Forest Algorithm for weather station : BUDAPEST



The Random Forest model has key benefits:

- Its ensemble nature reduces the risk of overfitting, especially compared to a single decision tree.
- It performs well even with large datasets and many input features.

Machine Learning Model 2

A generator network produces artificial data based on real data points. A discriminator network samples both and attempts to identify accurate data. The generator then uses these results to create more effective "fakes."

Convolution Neural Network (CNN)

A CNN uses hidden layers to automatically learn important patterns in the data and make classifications. Pooling layers simplify the data by keeping only key information, like averages or maximums, which helps reduce the amount of computation needed.

Benefits:

- Radar-generated graphics
- Analyzing complex data
- Parameter Efficiency

Thought Experiment #1: Classifying Weather Using Random Forest.

Hypothesis: A random forest model can identify abnormal weather trends based on historical patterns and station data.

Objective: Determine the safest places for people to live in Europe over the next 25 to 50 years

Approach:

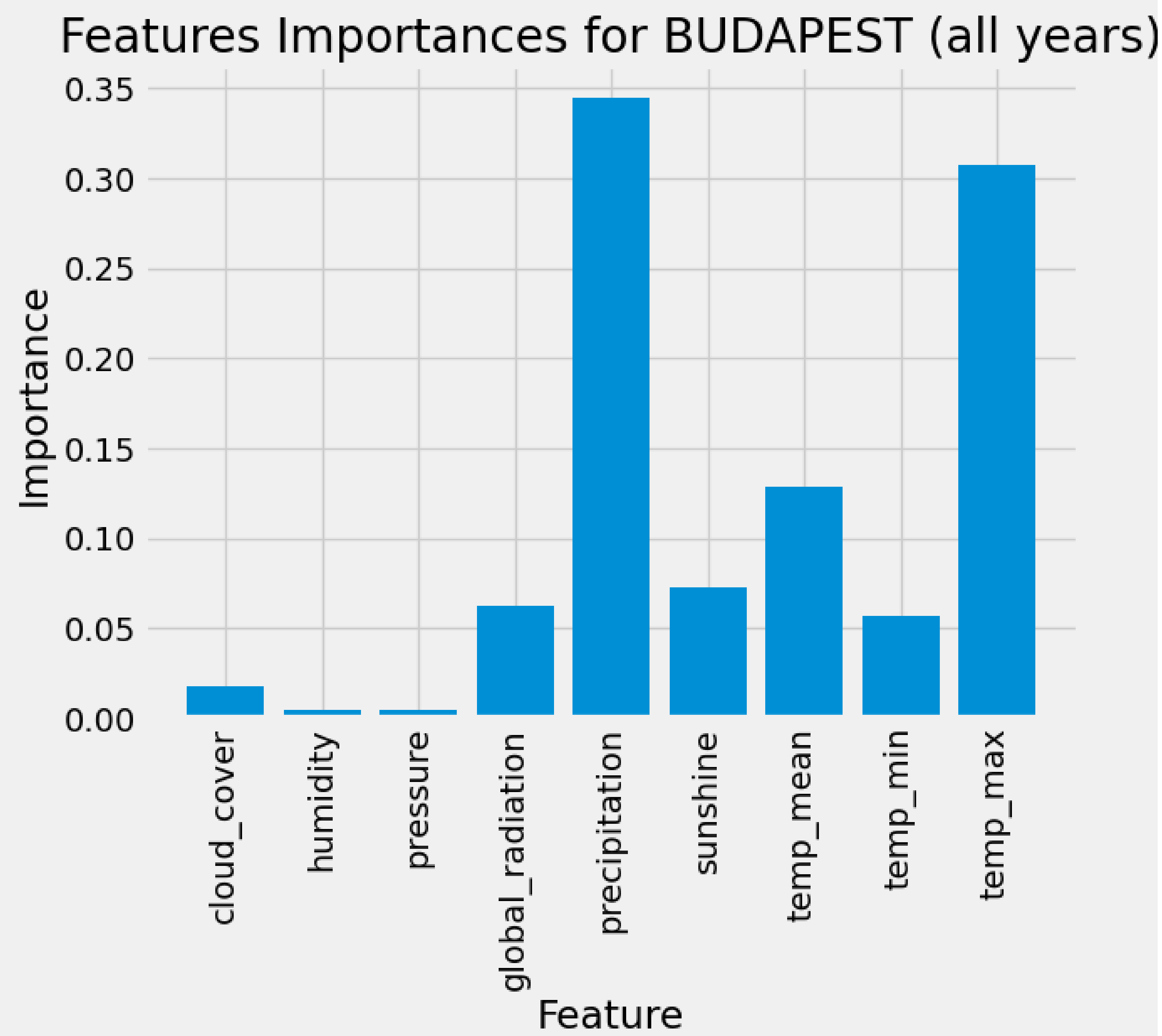
- Utilized the RandomForestClassifier to examine important factors like precipitation and temperature from multiple weather stations across Europe.
- Conducted hyperparameter tuning with RandomizedSearchCV to fine-tune settings such as n_estimators, max_depth, and min_samples_split, enhancing the model's overall accuracy.

Result:

- Accuracy: Improved from an initial 65% to approximately 73% after hyperparameter optimization.
- Feature Importance: The most predictive features were LJUBLJANA, MUNCHENB, and BUDAPEST stations, highlighting the areas with significant weather variation.

Thought Experiment #1: Classifying Weather Using Hierarchical Clustering

Conclusion: This approach successfully identified areas experiencing deviations from historical patterns, helping to detect increasing anomalies like shifts in precipitation and temperature extremes.



Thought Experiment #2: Classifying Weather Using Deep Learning for Image-Based Weather Classification.

Hypothesis: CNNs can better interpret radar and satellite imagery to classify weather conditions, improving the prediction of weather trends.

Objective: Assess climate risk across different regions in Europe by integrating image analysis, historical data, and predictive modeling.

Approach:

- Built a CNN model to categorize radar images based on different weather conditions such as cloudy, rainy, and sunny.
- Employed Bayesian optimization to fine-tune hyperparameters—including neuron count, batch size, and learning rate—to boost model performance and accuracy.

Result:

- Initial Accuracy: The unoptimized CNN achieved around 11% accuracy. After Bayesian optimization, accuracy improved significantly to 80%.
- Confusion Matrix: Showcases the model's ability to differentiate between weather conditions such as 'cloudy' and 'rainy,' highlighting areas where classification errors reduced after optimization.

Thought Experiment #2: Classifying Weather Using Deep Learning for Image-Based Weather Classification.

Model Used: CNN with Bayesian optimization.

Conclusion: This model demonstrated the potential for analyzing complex visual data, making it useful for predicting shifts in weather patterns. The optimization of the model led to a substantial improvement in performance, with the accuracy increasing by 80.28%, going from a relatively low 11.9% to a much higher 92.2%.

GANs for Synthetic Weather Projections

Hypothesis:

GANs can generate new weather scenarios based on current trends, providing a range of possible futures.

Approach:

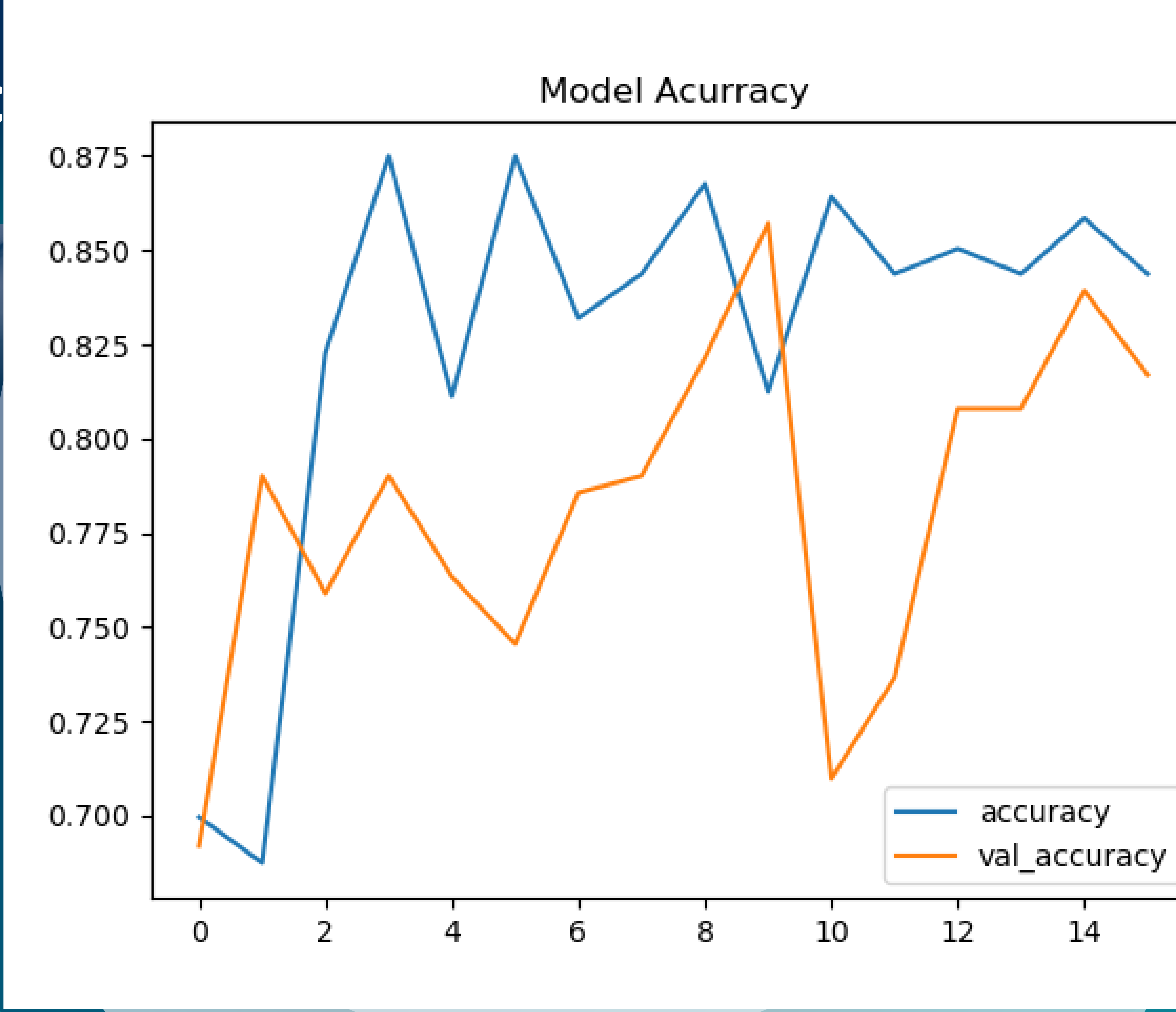
Use GANs to simulate various weather patterns and scenarios, such as temperature shifts or precipitation changes over time. Create synthetic weather maps to explore long-term predictions (e.g., 25 to 50 years).

GANs for
Synthetic Weather Project

Results:

Bayesian optimization improved the accuracy of CNN modeling from 10.4% to 61.2%.
The GAN produced weather data with 98.4% accuracy and 1.1% loss, suggesting high potential for use with the CNN model.

Conclusion: A tool that visualizes potential future climates, allowing stakeholders to plan for extreme scenarios.



Closing Discussion

Recommendation:

Use Random Forest models for immediate analysis of feature importance and to identify key predictors of abnormal weather patterns. Implement CNNs for analyzing satellite data and weather imagery to improve real-time classification of weather conditions. Invest in developing GANs for longer-term scenario simulation, helping ClimateWins plan for potential future climates.

Next Steps:

Validate model predictions with real-time data. Collaborate with meteorological agencies for improved data access. Explore scaling the models to include more complex climate data.

Keep Connected With Us

Our Email
GILLAMRIT03@OUTLOOK.COM

Website
https://github.com/gillamrit92

Our Location
Canada

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